ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

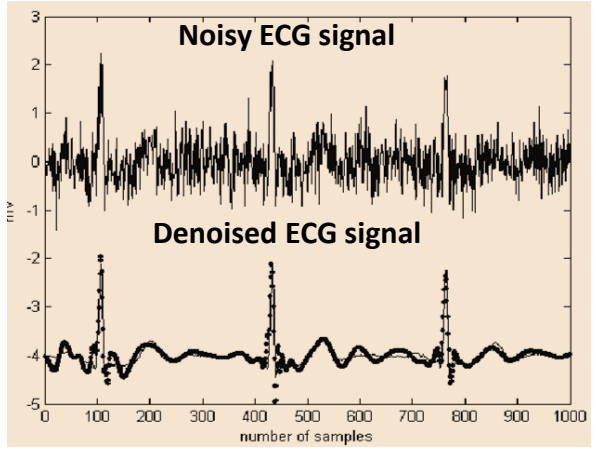
- Discrete-time signals & difference equations (pp. 3-11)
- Time and frequency domain analysis (pp. 12-15)
- Sampling theorem & aliasing (pp. 16-23)
- A/D conversion & quantisation errors (pp. 24-31)
- Tutorial questions (pp. 32-36)

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Term 3, 2021

Digital Signal Processing (DSP) Fundamentals

- In the 1990s the digital signal processing revolution started, both in terms of the consumer boom in digital audio, digital telecommunications and the wide used of technology in industry.
- The professional music industry uses spectrum analysers, digital filtering, sampling conversion filters etc and is one of the biggest users and exploiters of DSP technology. This technology is found inside headphone, smartphones, smart speakers and is actually a cornerstone of modern audio products..
- In the medical field, DSP systems are widely utilized for recording data analysis and the interpretation of ECG signals.
- Audiologists and speech therapists are exposed to DSP systems for both testing a person's level of hearing and subsequently DSP hearing aid filtering.
- DSP is applied in the area of control and power systems, biomedical engineering, instrumentation, automotive engineering, telecommunications, mobile communication, speech analysis and synthesis, audio and video processing, seismic, radar and sonar processing and neural computing.





Advantages using DSP

- High reliability and reproducibility
- High accuracy
- Flexibility and programmability
- Absence of component drift problem
- Compressed storage facility
- Cheaper

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Discrete-time signals & Difference equations

✓ There are **two** types of signals:

Continuous - time signals

A continuous-time signal, $x(t)$, the independent variable t is continuous and thus $x(t)$ is defined for all t $(-\infty < t < \infty)$ – *see Figure 1*

Discrete - time signals

- Discrete-time signals are defined only at discrete times and consequently the independent variable n takes on only a discrete set of values .
- A discrete- time signal is thus a sequence of numbers where n is a discrete time - independent variable ($n = \ldots$ -2, -1, 0, 1, 2,…) - *see Figure 1*

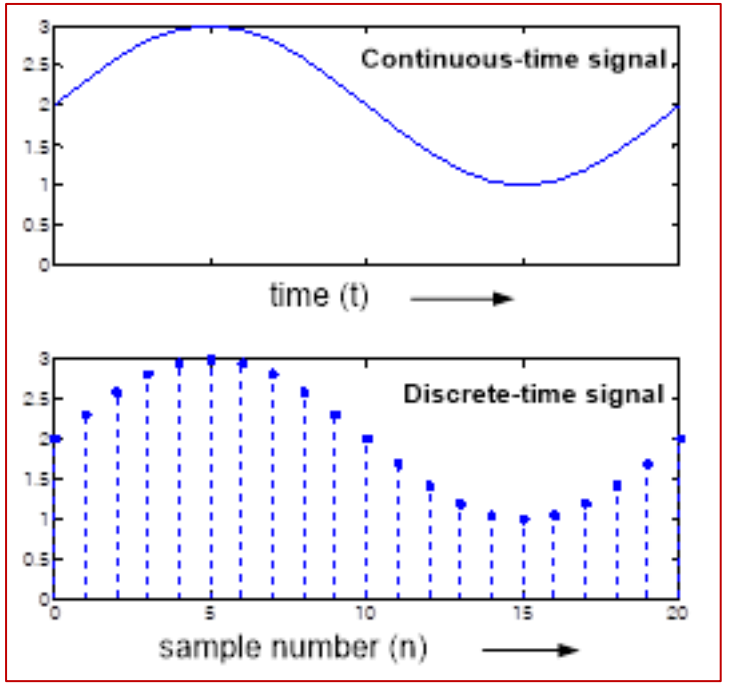


Figure 1: An example of continuous-time and discrete-time signals

Examples

- A person's body temperature is a continuous-time signal.
- The prices of stocks printed in the daily newspapers are discrete-time signals.
- Voltages and currents are usually represented by continuous-time signals. They are also represented by discrete-time signals if they are specified only at a discrete set of values of t.

Discrete-time signals: Sampling

Sampling

A discrete-time signal is often formed by sampling a continuous -time signal *x(t)*. If the samples are equidistant then;

$$x[n] = x(t)\big|_{t=nT} = x(nT)$$

Square brackets [ ] ⇒ Discrete time signals
Round Brackets ( ) ⇒ Continuous signals

The constant *T* is the sampling interval or period and $f_s=1/T$ is the sampling frequency.

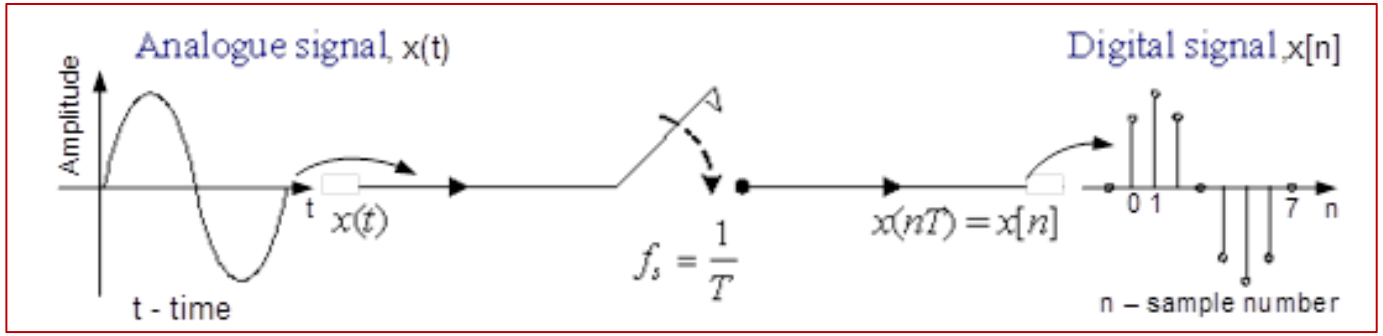
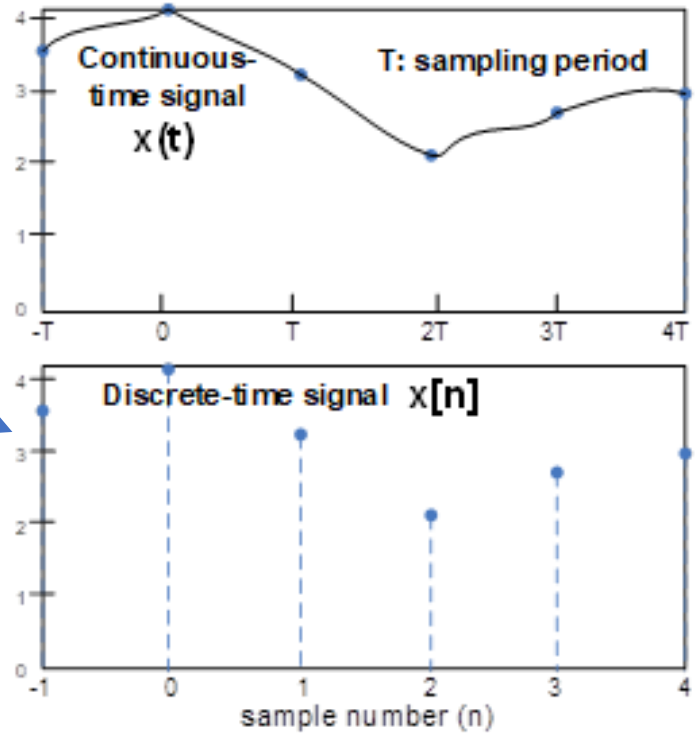


Figure 2: An example of acquiring discrete-time signal by sampling continuous-time signal

$$x[n] = \{\ 3.5,\ \ 4,\ \ 3.25,\ \ 2,\ \ 2.5,\ \ 3.0\ \}$$

n=-1 n=0 n=2 n=4

It is important to recognize that *x[n]* is only defined for integer values of *n*. It is not correct to think of *x[n]* as being zero for *n* not an integer, say *n* =1.5. *x[n]* is simply undefined for non-integer values of *n*.



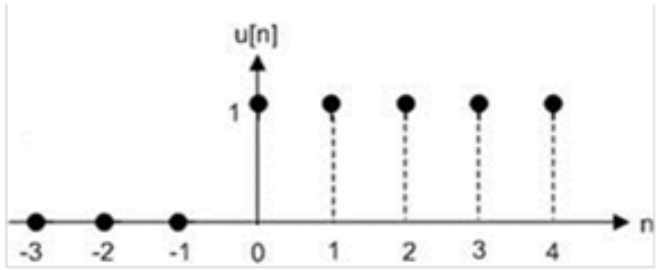
Sampling frequency $(f_s) = \frac{1}{T}$

Figure 3: An example of acquiring discrete-time signals by sampling continuous-time signals.

Discrete-time signals: Unit step, Unit impulse & Exponential

Unit step function

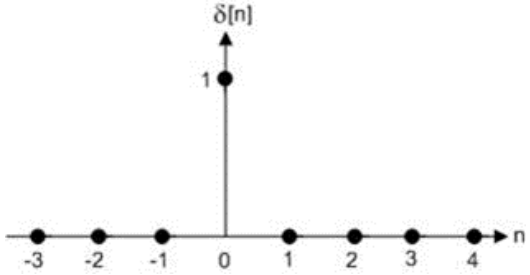
$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



The Unit Impulse

- An important concept in the theory of linear systems is the continuous time unit impulse function.
- The discrete-time unit impulse function $\delta[n]$ is defined in a manner similar to its continuous time counterpart. We also refer $\delta[n]$ as the unit sample

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



Discreate –time exponential signals

- In discrete time, it is common practice to write a real exponential signal as: $\boldsymbol{x[n] = c\,\alpha^{n}}$
- If c and α are real and if $|\alpha| > 1$ the magnitude of the signal grows exponentially with n, while if $|\alpha| < 1$, we have decaying exponential (see Figure 4)

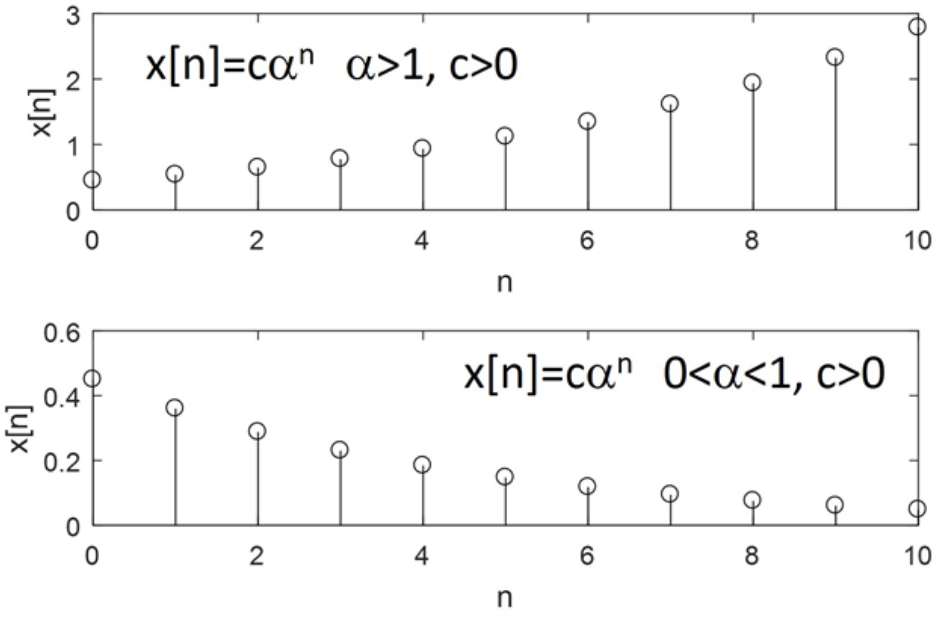


Figure 4: Examples of discrete-time exponential signals

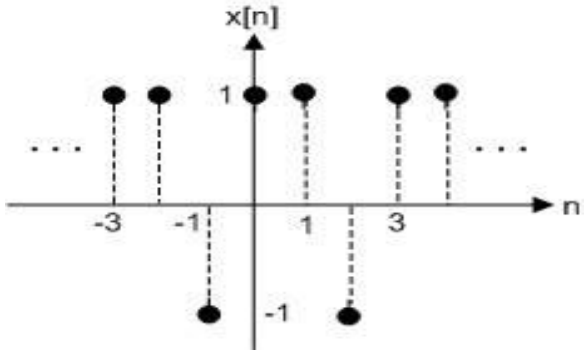
Discrete-time signals : Periodic signals

Periodic Signals

- A discrete-time signal *x[n]* is periodic with period *N*, where *N* is a positive integer, if for all values of *n*.

$$x[n] = x[n+N]$$

x[n] with Period, *N* = 3 samples



Discrete-time sinusoidal signals

- A continuous-time sinusoidal signal is given by:

$x(t) = A\sin(\omega_a t) = A\sin(2\pi f_a t)$ f_a: Analogue frequency

- A discrete - time sinusoidal signal may be expressed as:

$$x[n] = x(t)|_{t = nT} = x(nT)$$

$$x[n] = A\sin(n\boldsymbol{\omega_a}\mathrm{T}) = A\sin\left(2\pi\frac{f_a}{f_s}\mathrm{n}\right) = Asin(n\theta)$$

$$\boldsymbol{Digital\ frequency}\ (\theta) = \boldsymbol{2\pi\frac{f_a}{f_s}} = \boldsymbol{\omega_a T}\ \text{radians}$$

$$\boldsymbol{-\pi \le \theta \le \pi}$$

$$\boldsymbol{Sampling\ frequency}\ (f_s) = \frac{1}{T}\ \text{Hz}$$

- A discrete-time signal is said to be periodic with a period length *N*, if *N* is the smallest integer for which

$$x[n+N] = x[n]$$

$$\therefore A\sin((n+N)\theta) = A\sin(n\theta)$$

which can only be satisfied for all *n* if, $N\theta = 2\pi k$ (where *k* is an arbitrary integer); $N = \frac{2\pi k}{\theta} = \frac{2\pi k}{2\pi\frac{f_a}{f_s}}$ $\boxed{N = \frac{f_s}{f_a}k}$

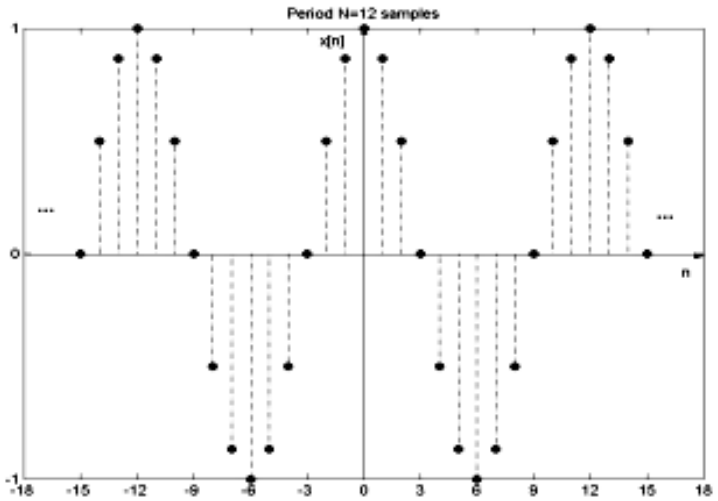
- For example, f_a = 1000 Hz and f_s = 8000 Hz then $N = f_s / f_a$ = 8 samples (assume *k* is the smallest positive integer, i.e. *k* = 1). Therefore the period length N = 8 samples.

Discrete-time signals: Examples

Example 1:

- The following diagram depicts the cosine wave *with* period *N* = 12 samples.

$$x[n] = \cos\left(\frac{2\pi n}{12}\right)$$



Example 2:

- Determine the fundamental period of $x[n] = 10\cos\left(\frac{2\pi}{15}n + \frac{\pi}{3}\right)$

$$Digital\ frequency\ (\theta) = 2\pi/15$$

Recall the equation: $N = \frac{2\pi k}{\theta}$; k =1 is the smallest positive integer.

$$\therefore N = \frac{2\pi.1}{2\pi/15} = 15 \text{ samples}$$

Example 3:

- The sinusoidal signal *x[n]* has fundamental period *N=10* samples. Determine the smallest θ for which *x[n]* is periodic:

$$\theta = {}^{2\pi k}/_{N} = {}^{2\pi k}/_{10};$$

Smallest value of θ is obtained when *k* =1;

$$\therefore \theta = \frac{2\pi}{10} = \frac{\pi}{5} \text{ radians}$$

Example 4:

- Determine if or not the discrete-time signal, *x[n]*, below is periodic and if it is periodic, determine the fundamental period. $x[n] = \sin[\frac{2}{10}\,n + \pi]$.

Ans: $x[n] = \sin[0.2\,n + \pi] = \sin[0.2\,(n + N) + \pi]$ where N is an integer period.

∴ 0.2 N = 2 π k (k is an arbitrary integer and sine function is periodic with a period of 2 π.

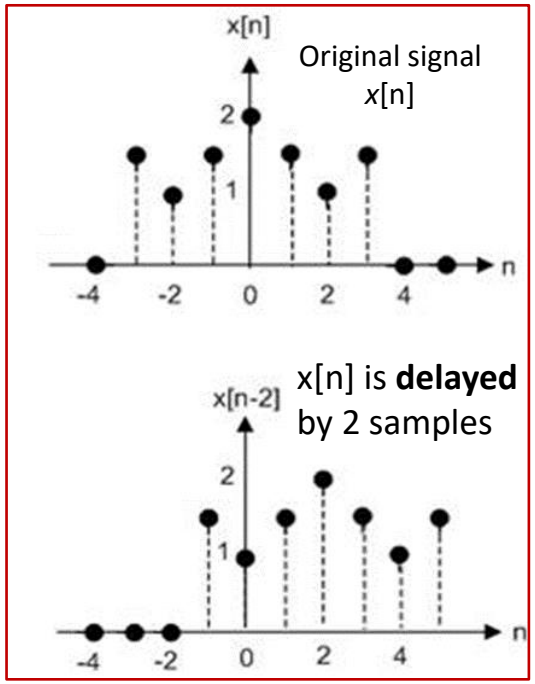
∴ N = 5 (2 π) k [non integer value of N exists]

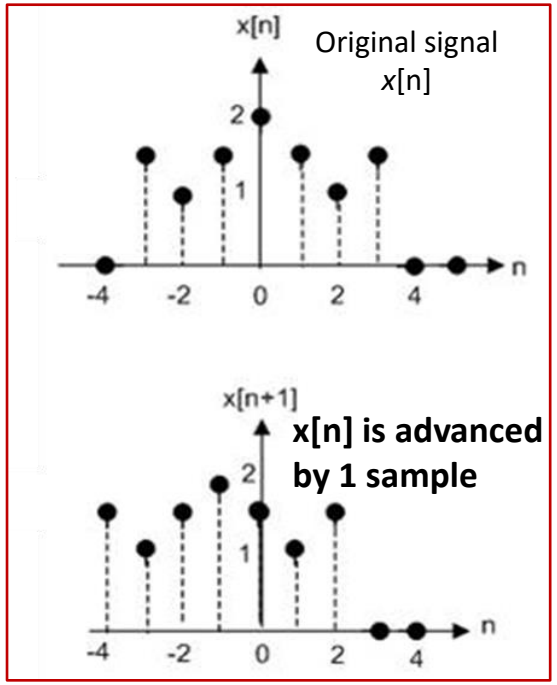
Since π is a irrational number (cannot be expressed as a ratio of two integers) no integer value of N exists. Therefore, the signal is aperiodic.

Discrete-time signals : shifted signals

Simple manipulation of discrete-time signals

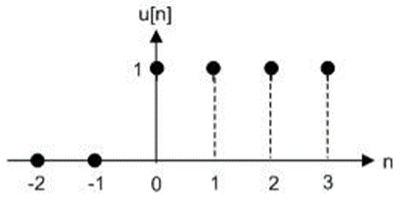
- A signal $x[n]$ may be shifted in time by replacing the independent variable n by $n-k$ where k is an integer.
- If $k > 0 \Rightarrow$ the time shift results in a **delay** of the signal by k samples [i.e. shifting a signal to the right]
- If $k < 0 \Rightarrow$ the time shift results in an **advance** of the signal by k samples. [i.e. shifting a signal to the left]



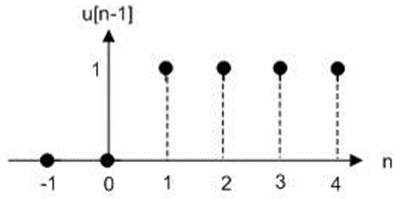


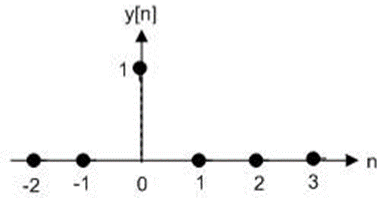
Example 5:

- Sketch the waveform : $y[n] = u[n] - u[n-1]$



$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$





Example 6:

- A continuous time signal is given by the following equation: $x(t) = 10e^{-t} - 5e^{-0.5t}$

  Write an equation for the discrete-time signal x[n] and find $x[n-2]$.

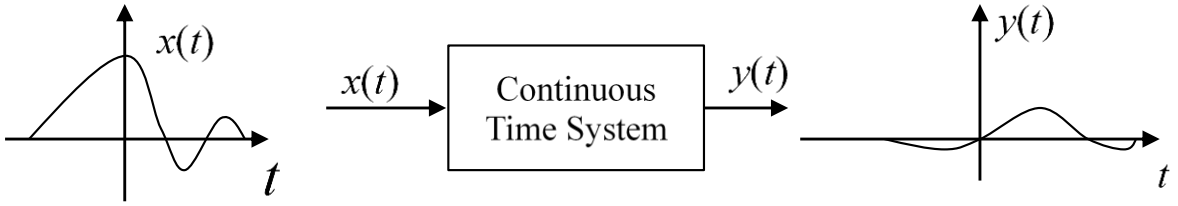
Ans: Substitute $t = nT$ in $x(t)$; [n - sample number & T – sampling period] $x[n] = 10^{-nT} - 5e^{-0.5nT}$

$$x[n-2] = 10^{-(n-2)T} - 5\,e^{-0.5(n-2)T}$$

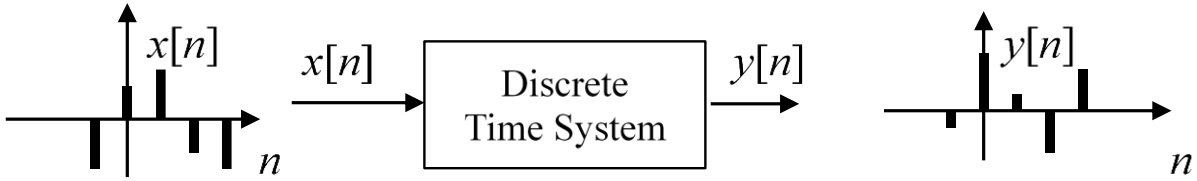
Difference equations

Systems

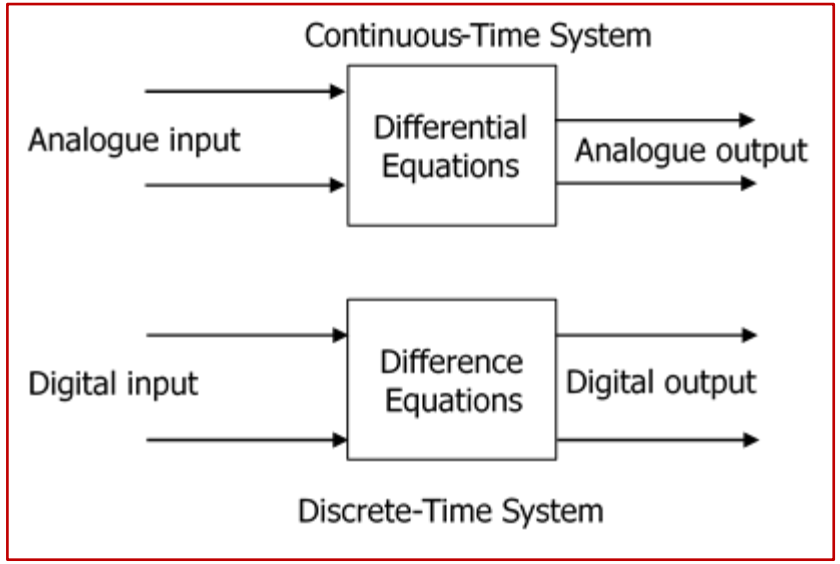
- A continuous-time system is one whose input *x*(t) and output *y*(t) are continuous time functions related by a rule as shown below:



- A discrete system is one whose input *x*[n] and output *y*[n] are discrete time function related by a rule as shown below:



- An important mathematical distinction between continuous-time and discrete-time systems is the fact that the former are characterized by differential equations whereas the latter are characterized by differen**ce** equations.



Difference equations

Example 7:

- The RC circuit (known as a low-pass filter or an integrator) shown below is a continuous-time system.
- If $e(t)$ is the input signal and $v_c(t)$ is the corresponding output signal, then we obtain using simple circuit analysis

$$\frac{dv_C(t)}{dt}+\frac{1}{RC}v_C(t)=\frac{1}{RC}e(t) \qquad (1)$$

Differential equation

- From equation (1), a discrete-time system can be developed. If the sampling period T is sufficiently small, by applying Backward Euler approximation , we obtain

$$\left.\frac{dv_C(t)}{dt}\right|_{t=nT}=\frac{v_C(nT)-v_C(nT-T)}{T} \qquad (2)$$

- By substituting equation (2) into (1) and replacing t by nT, we obtain

$$\frac{v_C(nT)-v_C(nT-T)}{T}+\frac{1}{RC}v_C(nT)=\frac{1}{RC}e(nT)$$

- The difference equation is:

$$\frac{v_C[n]-v_C[n-1]}{T}+\frac{1}{RC}v_C[n]=\frac{1}{RC}e[n]$$

Output

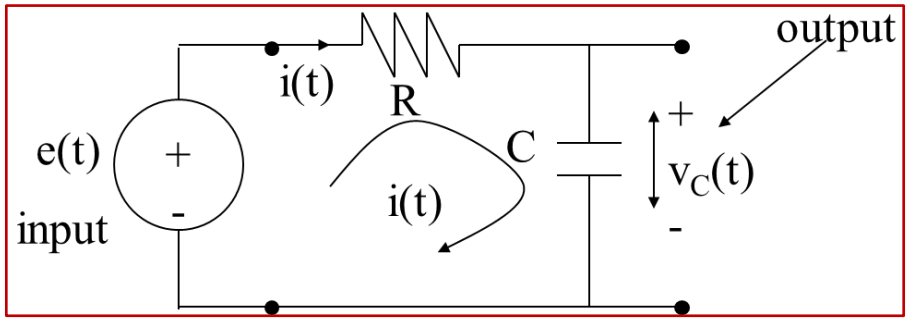
input

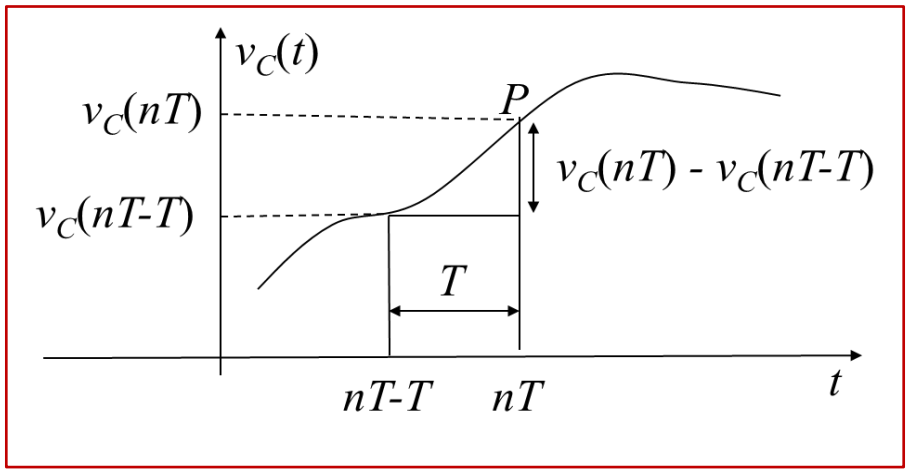
$$v_C[n]=\frac{RC}{RC+T}v_C[n-1]+\frac{T}{RC+T}e[n]$$

Previous output

Difference equation

RC circuit





Backward Euler approximation
[Assuming T is sufficiently small]

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Time and frequency domain analysis

Continuous-time Fourier Transform (FT)

- The Fourier transform for non-periodic continuous-time signals is a mathematical transformation to transform signals between time domain and frequency domain, which has many applications in engineering. Most signals of practical importance are non-periodic. The FT pairs is given by:

  Fourier Transform (FT): $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

  Inverse Fourier Transform (IFT): $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$

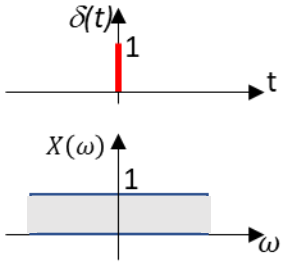
- The FT converts the time domain signal, *x*(t), into its frequency domain representation, $X(\omega)$, and the IFT converts the frequency domain representation, $X(\omega)$, back into the time domain *x*(t).

Example 8:

- Find the Fourier Transform of $x(t) = \delta(t)$.

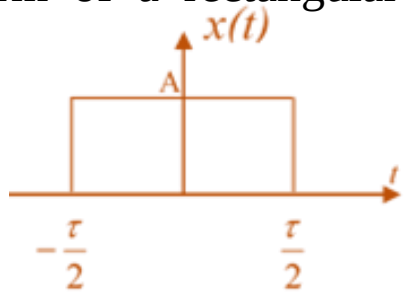
$$X(\omega) = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}dt = 1$$

- The unit impulse $\delta(t)$ contains all frequencies $(-\infty \ to \ +\infty)$ of unit amplitude.



Example 9:

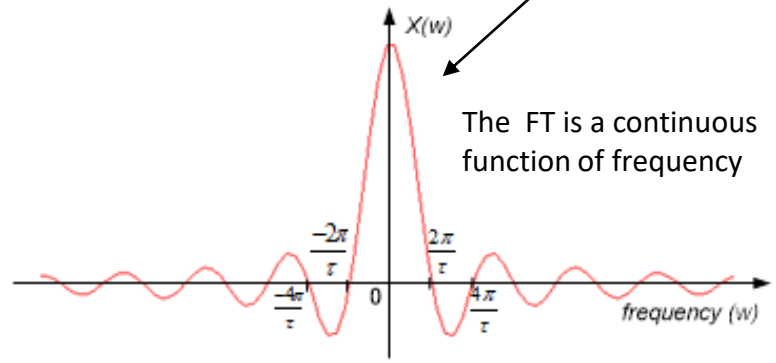
Evaluate the Fourier transform of a rectangular pulse shown below:



$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

$$= \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} x(t)e^{-j\omega t}\,dt \quad = -\frac{A}{j\omega}\left[e^{-\frac{j\omega\tau}{2}} - e^{\frac{j\omega\tau}{2}}\right]$$

$$X(\omega) = A\tau \frac{sin\frac{\omega\tau}{2}}{\frac{\omega\tau}{2}}$$ ← Sinc function



The FT is a continuous function of frequency

Time and frequency domain analysis

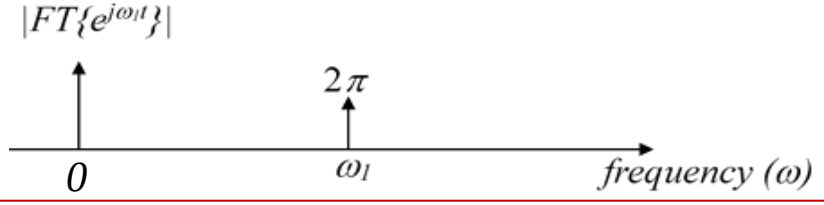
Example 10:

- Find the Fourier transform of $x(t) = e^{j\omega_1 t}$.

$$FT\{e^{j\omega_1 t}\} = \int_{-\infty}^{\infty} e^{j\omega_1 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{j(\omega_1 - \omega)t} dt = 2\pi\delta(\omega_1 - \omega)$$

$$FT\{e^{j\omega_1 t}\} = 2\pi\delta(\omega - \omega_1)$$

Hence the magnitude spectrum of $e^{j\omega_1 t}$ is shown below:



Note: Use the following properties:

$\int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi\delta(\omega)$

and

$\delta(\omega - \omega_1) = \delta(\omega_1 - \omega)$

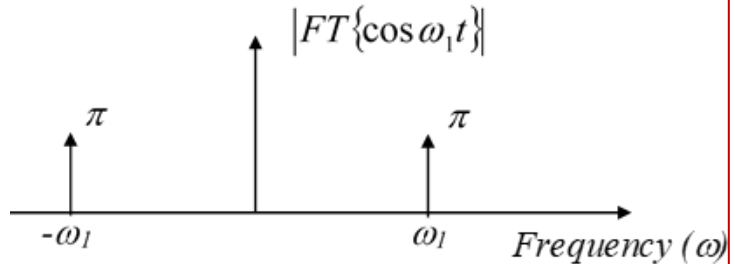
Example 11:

- Find the Fourier transform of $x(t) = \cos(\omega_1 t)$.

$$FT\{\cos\omega_1 t\} = FT\left\{\frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2}\right\} = \frac{1}{2}FT\{e^{j\omega_1 t}\} + \frac{1}{2}FT\{e^{-j\omega_1 t}\}$$

$$= \pi\delta(\omega - \omega_1) + \pi\delta(\omega + \omega_1)$$ Use results from Example 10.

The magnitude spectrum of $\cos(\omega_1 t)$ is shown below:



- The spectrum of cosine signal has two impulses at positive and negative frequencies.

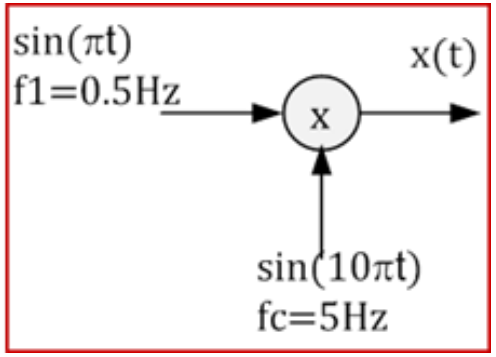
Time and frequency domain analysis

Multiplication of sinusoids

- The product of two real sinusoids (frequencies f_1kHz and f_2kHz) results in the sum and difference frequencies $[(f_1+f_2)$kHz and (f_1-f_2)kHz$]$

- Consider a signal $x(t)$ which is a product of two sinusoids, as given by:

$$x(t) = \sin(\pi t) \text{ x } \sin(10\pi t)$$



$x(t)$ can be re-written as

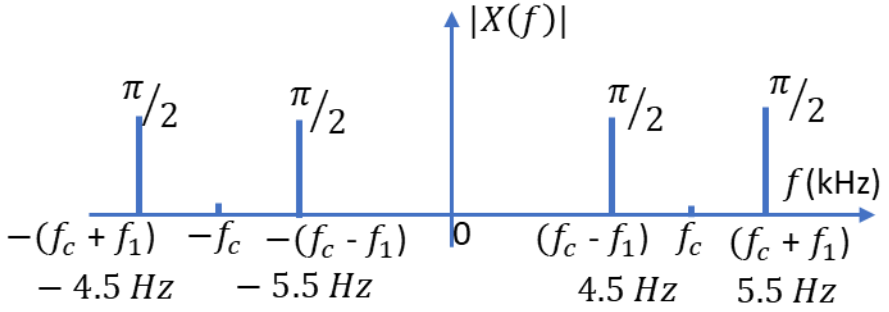
$$x(t) = \left(\frac{e^{j\pi t} - e^{-j\pi t}}{2j}\right)\left(\frac{e^{j10\pi t} - e^{-j10\pi t}}{2j}\right) = \frac{-1}{4}\left(e^{j\pi t} - e^{-j\pi t}\right)\left(e^{j10\pi t} - e^{-j10\pi t}\right)$$

$$x(t) = \frac{-1}{4}[e^{j11\pi t} - e^{-j9\pi t} - e^{j9\pi t} + e^{-j11\pi t}] = \frac{1}{2}\cos(9\pi t) - \frac{1}{2}\cos(11\pi t) = \frac{1}{2}\cos(2\pi 4.5t) - \frac{1}{2}\cos(2\pi 5.5\,t)$$

- Recall example 11: $FT\{\cos\omega_1 t\} = \pi\delta(\omega-\omega_1)+\pi\delta(\omega+\omega_1)$

- From $x(t)$ it can be seen that there are four spectral components at frequencies f_c+f_1= 5.5Hz, f_c-f_1 = 4.5Hz, $-(f_c+f_1)$ = 4.5Hz, $-(f_c-f_1)$ =-5.5Hz, as shown below

Magnitude Spectrum of $x(t)$



ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Discrete-time signals & difference equations (pp. 3-11)
- Time and frequency domain analysis (pp. 12-15)
- Sampling theorem & aliasing (pp. 16-23)
- A/D conversion & quantisation errors (pp. 24-31)
- Tutorial questions (pp. 32-36)

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Term 3, 2021

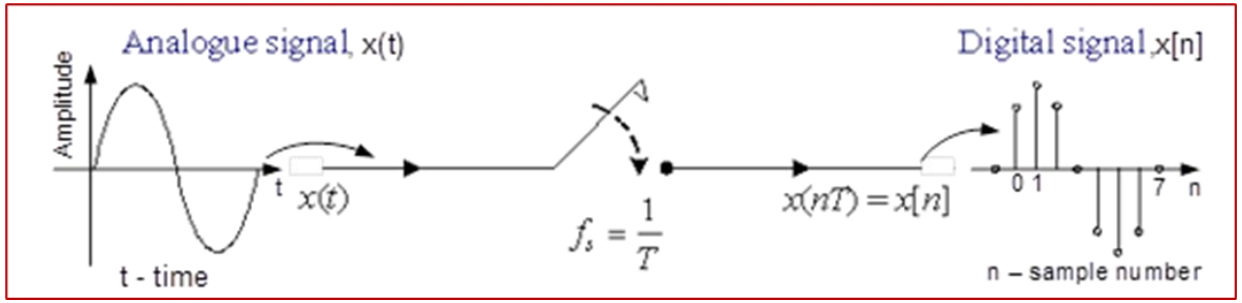
Sampling Theorem & Aliasing

Nyquist –Shannon Sampling theorem

- If the highest frequency contained in an analogue signal $x(t)$ is f_{max} and the signal is sampled at a rate $f_s \geq 2\, f_{max}$ then $x(t)$ can be exactly recovered from its sample values using an interpolation function.
- The maximum frequency (f_{max}) that is generally accepted as upper limit of human hearing and perception of music sounds is 20kHz.
- Audio CDs use a sampling rate $f_s = 44.1kHz$ for storage of the digital audio signal. This sampling frequency is slightly more than $2 \times f_{max}$ [f_{max} = 20kHz] to avoid aliasing.

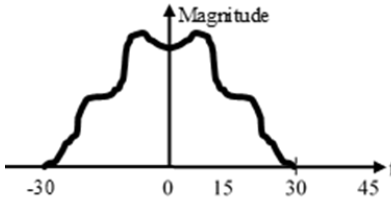
- For example, a sinewave of 50Hz (shown on the right) need to be sampled at higher than 100 Hz (i.e. 100 samples per second) in order NOT to lose any data. That is, to be able to reconstruct the 50Hz sinewave exactly.

Acquiring discrete-time signal, x[n] by sampling continuous-time signal x(t).

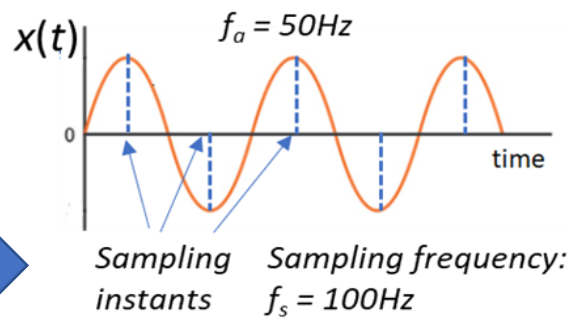


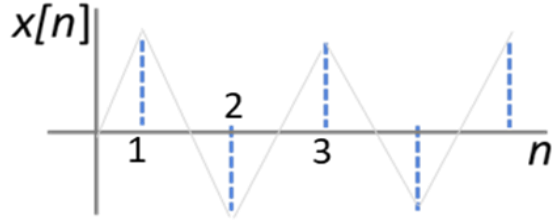
Example 12:

- What is the minimum sampling frequency, f_s, required to avoid aliasing when sampling a signal with the following spectrum?



a) 30 kHz
b) 60 kHz
c) 45 kHz
d) None of the above





Sampling Theorem & Aliasing

Overview of a DSP system

- An analogue signal processing system is shown Figure 5, in which both the input signal and output signal are in analogue form.
- A digital signal processing system in Figure 6 provides an alternative method for processing the analogue signal.

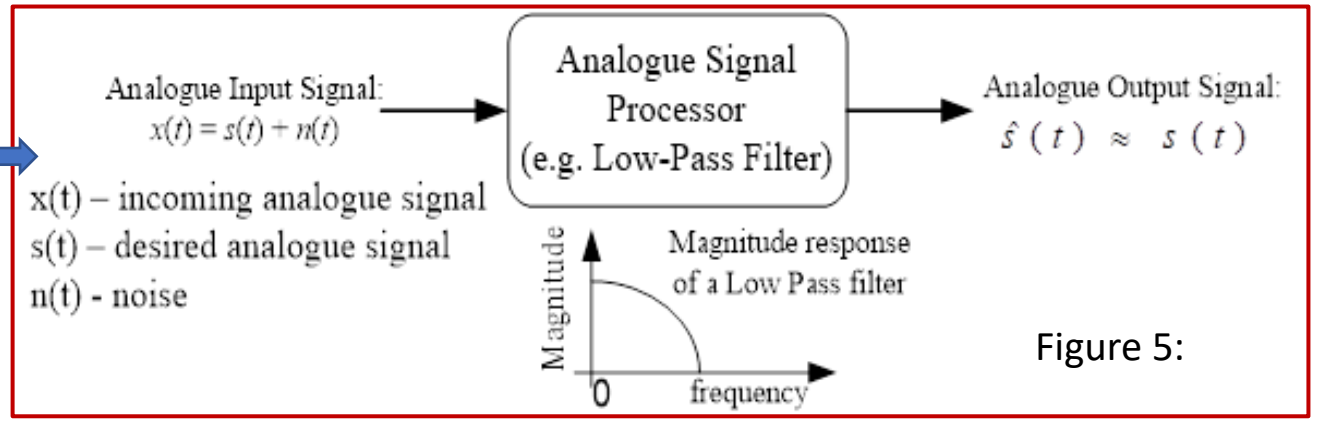


Figure 5:

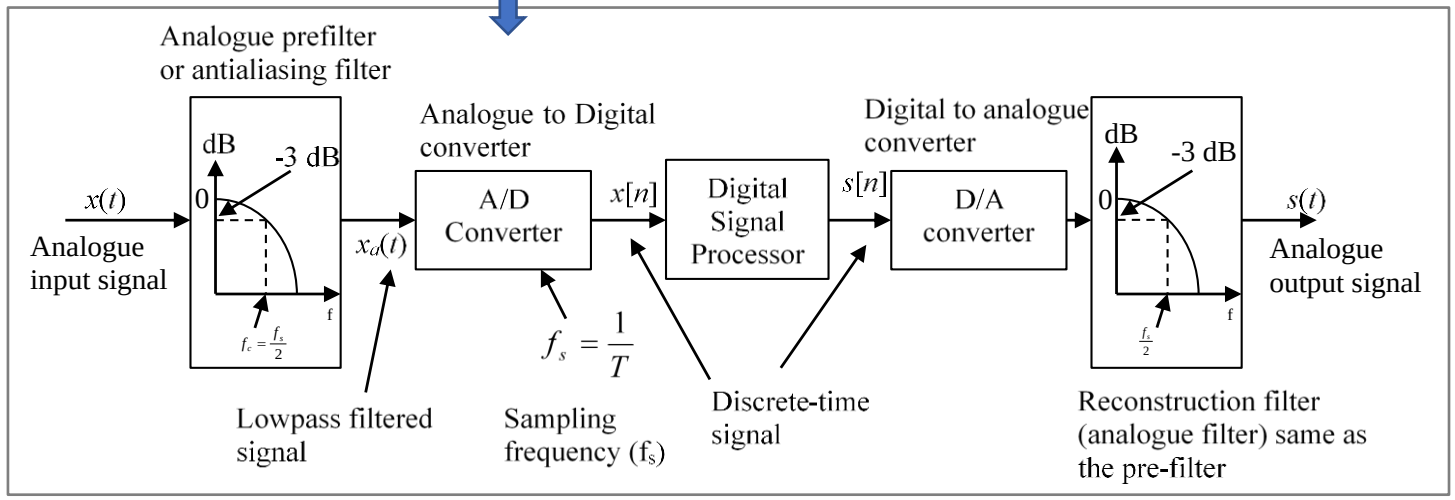


Figure 6: A general process of converting analogue signals into digital signals and back to analogue form.

The **digital signal processor** may implement one of the several DSP algorithms, for example digital filtering (low-pass filter) mapping the digital input signal $x[n]$ into digital output signal $s[n]$.

Name of the Block	Function
Anti-aliasing filter ($f_c = \frac{f_s}{2}$)	To band-limit the analogue input signal prior to digitisation to reduce aliasing
Analogue-to-digital converter (A/D)	To convert analogue input signal into digital output signal by sampling ($f_s = 1/T$)
Digital Signal Processor (heart of the system)	To process the digital signal according to the pre-defined rules.
Digital-to-analogue converter (D/A)	To convert the digital input signal into analogue output signal by interpolating.
Reconstruction filter ($f_c = \frac{f_s}{2}$)	To smooth out the output of D/A converter and to remove unwanted high frequency components.

Table 1: Description of blocks contained in a general DSP process

Sampling Theorem & Aliasing

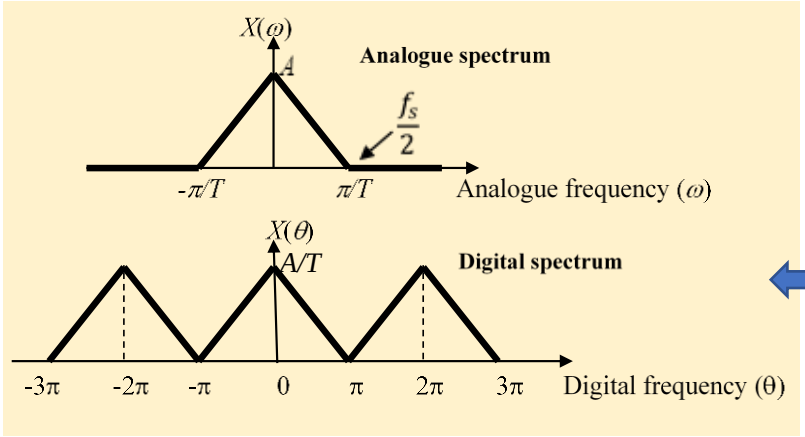
Sampling of Continuous –time signal

- If a continuous-time signal *x*(t) is sampled every *T* seconds, then at the output of the Analogue to Digital Converter, as shown below we obtain a discrete-time signal x[n]



- where t = time; n = sample number; ω = analogue frequency; θ = digital frequency $(-\pi \leq \theta \leq \pi)$; $\theta = \omega T$, $\theta = 2\pi\frac{f_a}{f_s}$, $X(\theta)$ = digital spectrum; $X(\omega)$ = analogue spectrum.

Note:
$\omega = \frac{\pi}{T}$ (or $f = \frac{f_s}{2}$) corresponds to $\theta = \pi$



Discrete-Time Fourier Transform Pair

- The Discrete-Time Fourier Transform (DTFT) pair is given by:

Discrete Time Fourier Transform (DTFT):
$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-jn\theta}$$

Inverse Discrete Time Fourier Transform (IDTFT):
$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{jn\theta} d\theta$$

$$where \ -\pi \leq \theta \leq \pi, \quad \theta = \omega T, \quad \theta = 2\pi\frac{f_a}{f_s}$$

- Using Inverse Fourier Transforms in Continuous-time and discrete-time domains, we can show that [see Topic 1: Tutorial question 3 (SOLO – Level 3)].

$$X(\theta) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T}) \qquad (3)$$

- It is seen that the digital spectrum $X(\theta)$ is periodic with period 2π. The digital spectrum is a repetition of the bandlimited analogue spectrum.

Sampling Theorem & Aliasing

Aliasing

- Aliasing arises when a continuous-time signal is sampled at a rate that is insufficient to capture the changes in the signal. If aliasing occurs, the original continuous time signal cannot be recovered.
- The Nyquist sampling theorem states that the sampling frequency, f_s, should be at least twice the highest frequency, f_c, contained in the signal ($f_s \geq 2f_c$) to avoid aliasing.
- Figure 7 illustrates the relationship between the digital spectrum $X(\theta)$ and the analogue spectrum $X(\omega)$ for the case $X(\omega) = 0$ where $|\omega| \geq \frac{\pi}{T}$ (i.e. $|f| \geq \frac{f_s}{2}$)

Case 1: $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T}$; (Sampling theorem holds)

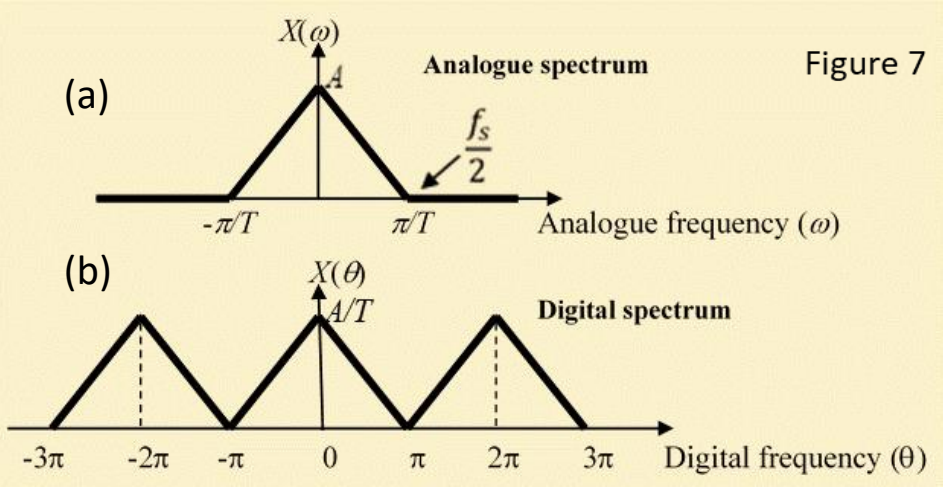


Figure 7

The digital spectrum is the same as the original analogue spectrum and repeats at multiples of the sampling frequency f_s

Case 2: $X(\omega) = 0$, $|\omega| \geq \frac{3\pi}{2T}$; (Sampling theorem does not hold)

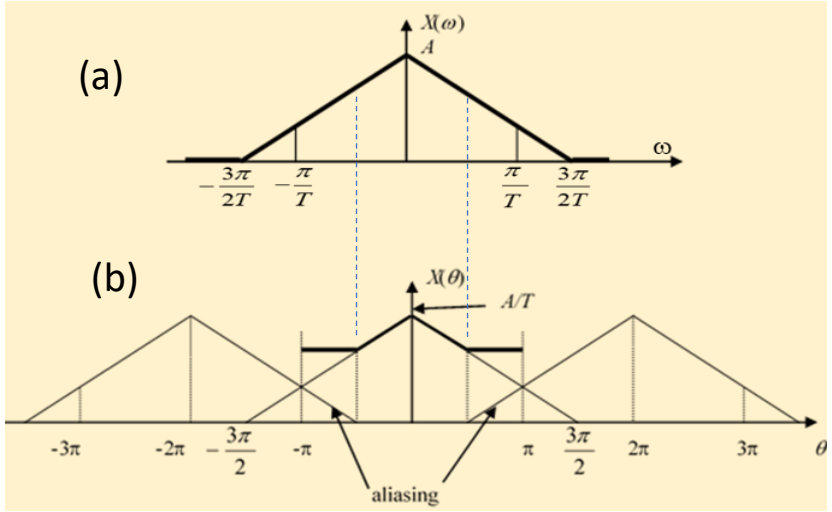


Figure 8: (a) Magnitude response of an analogue signal whose highest frequency component is greater than the sampling frequency. (b) Magnitude response of the sampled analogue signal. The overlapped region represents aliasing.

- If the sampling frequency, f_s is not sufficiently high, the spectrum centered on f_s will fold over or alias into the base band frequencies (Figure 8).
- Aliasing can only be avoided if the analogue signal is band limited such that $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T}$; [i.e. $|2\pi f| \geq \frac{\pi}{T} \Rightarrow \boldsymbol{f \geq \frac{f_s}{2}}$]

Sampling Theorem & Aliasing

Aliasing

- The digital spectrum consists of original band limited analogue spectrum and repeats at multiples of the sampling frequency f_s (refer to Figure 9) as given by:

$$f_k = f_0 + k f_s \qquad (4)$$

- where k is an integer ($k = \pm1, \pm2, \pm3, \ldots$) and f_0 is the frequency present in the fundamental region of the original analogue spectrum.
- It is clear that the frequency f_k is outside the fundamental frequency range

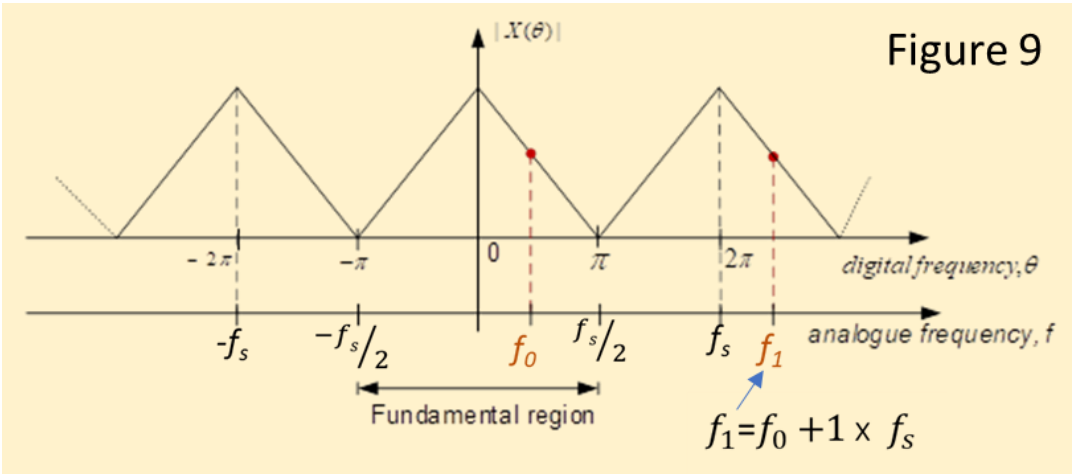
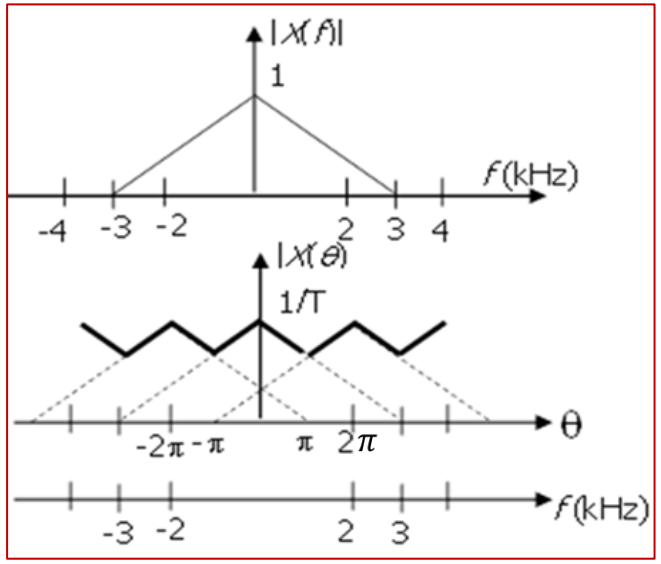


Figure 9

Example 13:

- Suppose $x(t)$ has the magnitude spectrum $|X(f)|$ as shown as shown on the diagram. Sketch the digital spectrum $|X(\theta)|$ if the sampling frequency f_s = 2 kHz.



Example 14:

- Consider the following analogue signal:

$$x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t)\ ;$$

What is the Nyquist rate for this signal?

Ans: The frequencies present in the signal above are:

$f_1 = 25$Hz; $f_2 = 150$ Hz; $f_3 = 50$ Hz; Hence, $f_{max} = 150$ Hz

$f_{sampling} > 2 f_{max} = 300$ Hz;

The Nyquist rate is $f_N = 2 f_{max} = 300$ Hz.

Sampling Theorem & Aliasing

Example 15:

- Consider the following analogue signal

$$x(t) = 3\,cos(2000\pi t) + 5\,sin(6000\pi t) + 10\,cos(12000\pi t)$$

(a) What is the Nyquist rate for this signal?

The frequencies existing in the analogue signal are:

f_1 = 1 kHz; f_2 = 3 kHz; f_3 = 6 kHz

Thus f_{max} = 6 kHz and according to the sampling theorem,

$f_s > 2\,f_{max}$ = 12 kHz

The Nyquist rate is = 12 kHz.

(b) Assume now that we sample this signal $x(t)$ using a sampling rate f_s = 5 kHz (samples/sec). What is the discrete-time signal obtained after sampling?

$$f_s = 5kHz \Rightarrow \frac{f_s}{2} = 2.5kHz$$

Recall equation (4): $\boldsymbol{f_k = f_0 + kf_s}$ [f_0 is the frequency present in the fundamental region of the original analogue spectrum] . Using equation (4), we can calculate the new frequency ($\boldsymbol{f'_n}$)caused by aliasing effect as follows:

$$\boldsymbol{f'_n = f_n - kf_s} \qquad -\frac{f_s}{2} \le \boldsymbol{f'_n} \le \frac{f_s}{2}; \qquad (k = \pm1, \pm2, \ldots)$$

$\boldsymbol{f'_n}$ - new frequency due to aliasing effect; $\boldsymbol{f_n}$ – frequency causing the aliasing (which is above the folding frequency)

- The frequency f_1 = 1000 Hz is $< \frac{f_s}{2}$ (= 2500 Hz) and thus it is not affected by aliasing.
- However, the other two frequencies f_2 & f_3 are above the folding frequency and they will be changed by the aliasing effect.

$$f_2' = f_2 - 1\,f_s = 3000 - 5000 = -2 \text{ kHz}$$

$$f_3' = f_3 - 1\,f_s = 6000 - 5000 = 1 \text{ kHz}$$

Therefore,

$$x[n] = 3\,cos\left(2\pi\left(\frac{1000}{5000}\right)n\right) + 5\,sin\left(2\pi\left(-\frac{2000}{5000}\right)n\right) + 10\,cos\left(2\pi\left(\frac{1000}{5000}\right)n\right)$$

(C) What is the analogue signal $y(t)$ we can reconstruct from the samples if we use ideal interpolation?

Since only frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analogue signal we can recover is,

$$y(t) = 13\cos(2000\pi t) - 5\sin(4000\pi t)$$

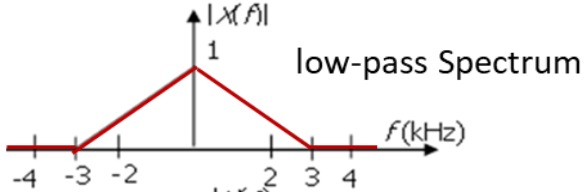
which is obviously different from the original signal x(t).

The distortion of the original analogue signal was caused by the aliasing effect, due to the low sampling rate used.

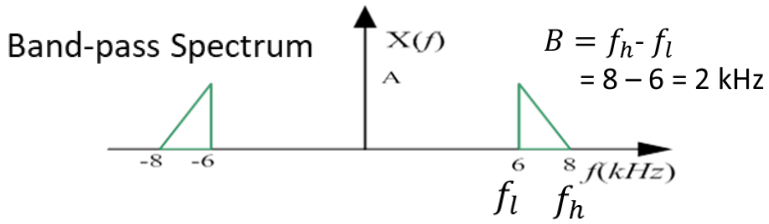
Sampling Theorem & Aliasing

Sampling Bandpass Signal

- The sampling of low-pass signals (see diagram below), is mainly the sampling scheme used in practice.



- We can use a technique known as **bandpass sampling** to sample a continuous bandpass signal that is centered about some frequency other than zero Hz (see diagram below).

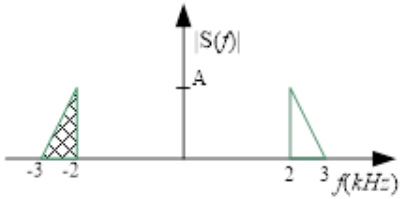


- The minimum sampling frequency (f_s) for a bandpass signal (see above) with a bandwidth (B) of 2 kHz is only about 4kHz ($f_s = 2\,B$), as opposed to twice the highest frequency of 16 kHz.
- $f_s = 2\,B$ holds (no aliasing will occur) only if f_h is an **integer** multiple of the Bandwidth *B*.

Example 16:

- Consider a bandpass signal, *s(t)*, with magnitude spectrum satisfying the following equation

$$|S(f)| = \begin{cases} 3 - |f| & 2\text{kHz} < |f| < 3\text{kHz} \\ 0 & \text{otherwise} \end{cases}$$



f – frequency in kHz. The signal *s(t)* is sampled uniformly with a sampling frequency of 2kHz. Sketch the digital spectrum $|S(\theta)|$, if it is sampled at a sampling frequency of 2kHz.

Ans: We have $f_s = 2$ kHz, and $f_k = f_o + kf_s$

[where *k* is an integer (*k* = ±1, ±2, ±3, …) and f_0 is the frequency present in the fundamental region of the original analogue spectrum]

The new frequency due to aliasing is: f_o'

$f_o' = f_o - kf_s, \qquad k = 0, \pm1, \pm2, \ldots$

For $k = 1$:

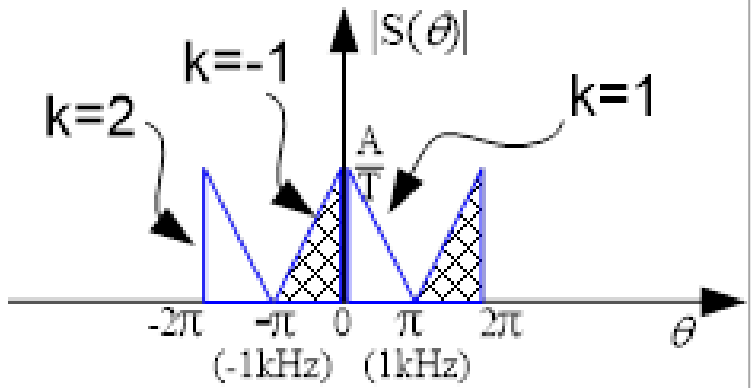
$f_o' = 2 - 1(2) = 0;$

$f_o' = 3 - 1(2) = 1$

For $k = -1$:

$f_o' = -2 + 1(2) = 0;$

$f_o' = -3 + 1(2) = -1$



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A/D Conversion & quantisation errors

Analogue to Digital (A/D) Conversion Process

- Before any DSP algorithm can be performed, the signal must be in a digital form [$x_a(t)$ to x[n] transformation – see Figure 10.].
- The A/D conversion process involves the following steps:
  - The signal (Band-limited) is first sampled, converting the analogue signal into a discrete-time signal
  - The amplitude of each sample is quantised into one of 2^B levels (where B is the number of bits used to represent a sample in the A/D converter)
  - The discrete amplitude levels are represented or encoded into distinct binary words each of length B bits.
- A practical representation of the A/D conversion process is shown in Figure 11.
- Sample and hold (S/H) takes a snapshot of the analogue signal every T sec and then holds that value constant for T secs until the next snapshot is obtained.

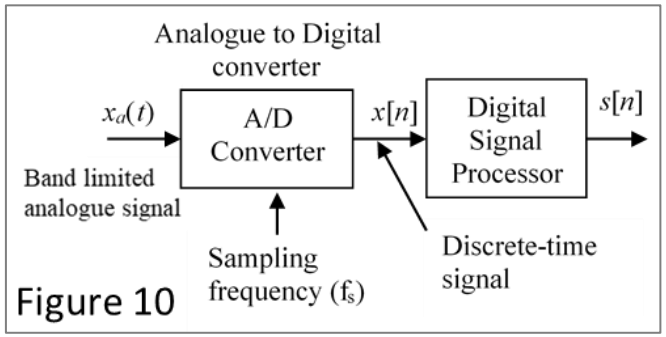


Figure 10

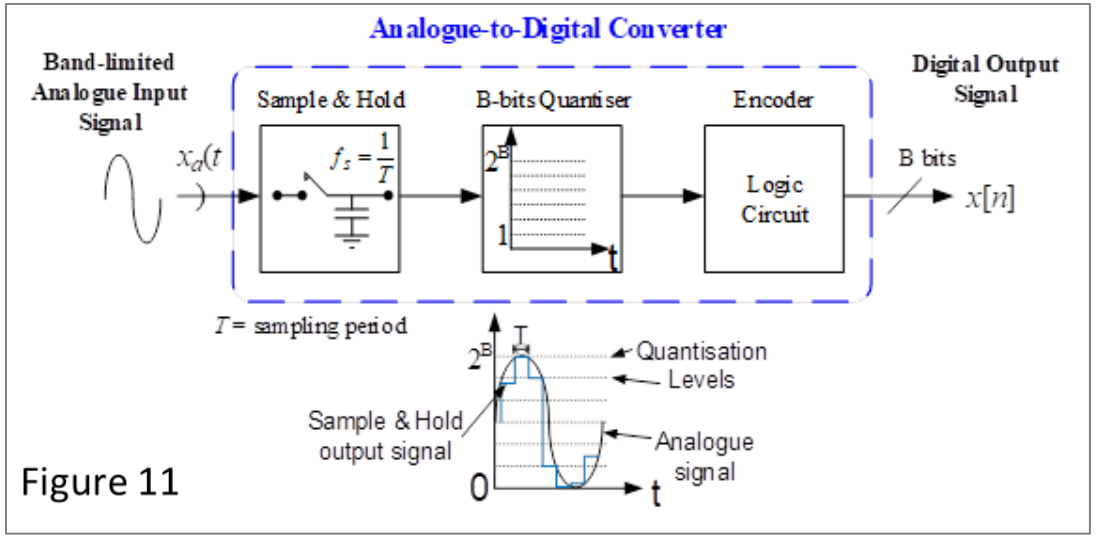


Figure 11

A/D Conversion & quantisation errors

Quantisation and Encoding

- Quantizing/encoding is the process of mapping the sampled analog voltage values to discrete voltage levels, which are then represented by binary numbers (B bits).
- Before conversion to digital, the analogue sample is assigned one of 2^B values (see Figure 12: **unipolar** 3 bit quantiser – 8 discrete levels). This process, termed quantization, introduces an error, which cannot be removed.
- A/D systems are characterized by the number of bits (e.g. 8-bit A/D; 12-bit A/D, 16-bit A/D) they have available to perform quantization. The number of bits determines the number of quantization levels.
- Input-output characteristics of a **bipolar** 4-bit A/D converter, with linear quantisation (two's complement notation) is shown in Figure 13.

An 8-level unipolar A/D converter coding scheme

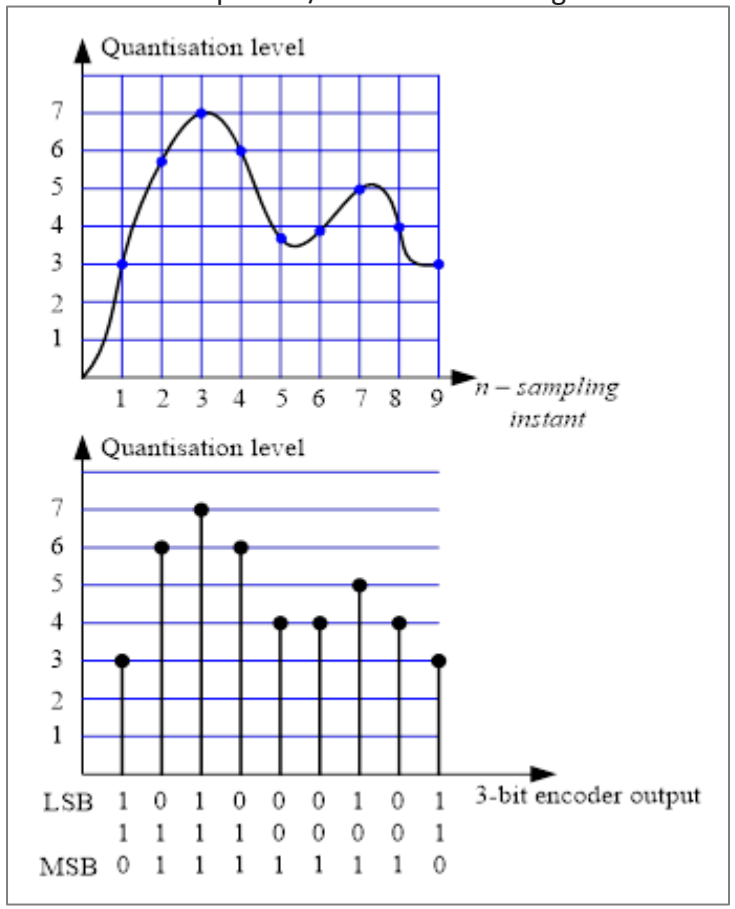


Figure 12: Quantisation of discrete-time signals

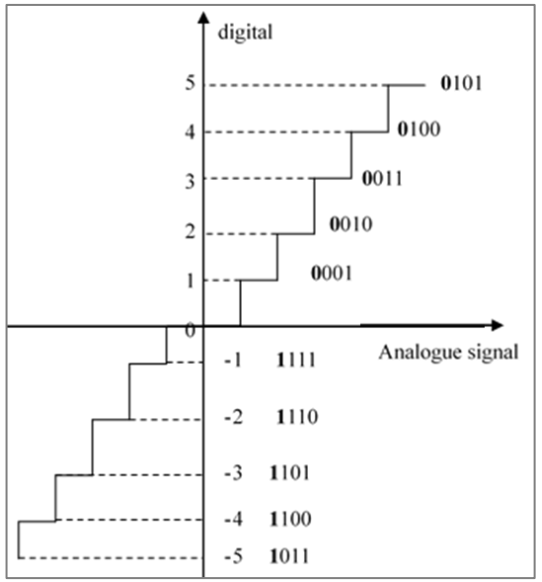


Figure 13: A 4 bit bipolar A/D converter with 2's complement coding scheme

Quantisation errors

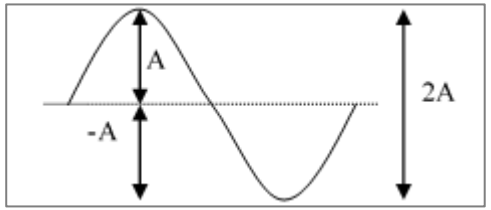
Quantisation step size

- For a B-bit A/D converter, the number of quantisation level is 2^B, and the interval between levels, that is known as the quantisation step size or resolution (ΔV) is given by:

  $\Delta V = \frac{V}{2^B - 1} \approx \frac{V}{2^B}$ [The approximation holds when B is Large (B >10 bits)

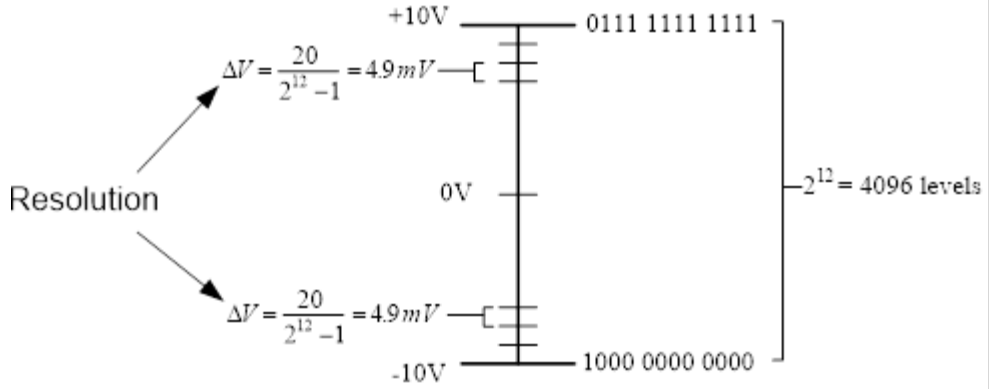
- where ΔV = resolution, B = number of bits and V = peak-to-peak amplitude.

- For a sine wave input of amplitude, A, the quantisation step size becomes: $\Delta V \approx \frac{2A}{2^B}$



Example 17:

- A 12 bit A/D converter (bipolar) with an input voltage range of ±10V will have a least significant bit (LSB) of $\frac{20V}{2^{12}-1} mV = 4.9mV$ (resolution or quantisation step size, ΔV)



Quantisation errors

Quantisation error (e)

- There is always error introduced with the A/D conversion process.
- The quantisation error, e[n], is the difference between the actual amplitude of the input signal and the quantised amplitude: $e[n] = A - A_q$

Actual amplitude — Quantized amplitude

- Figure 14 shows a portion of a signal that has been quantized with N bits. The green plot is the quantization error e[n]. The quantization error manifests as noise (it is also called quantization noise) in the reconstructed analogue signal.

- Quantization error can be reduced by increasing the number of bits N for each sample. This will make the quantization intervals smaller, thus reducing the difference between the input analogue signal and the quantization levels.

- For an A/D converter with bipolar signal inputs, the quantisation error (e[n]) is rounded up or down $\pm\ \Delta V/2$ (see Figure 15). The quantisation error ($e[n]$) for each sample, is normally assumed to be random and uniformly distributed in the interval $\pm\ \Delta V/2$ with zero mean.

- We can't use an infinite number of bits for the A/D conversion, so some quantization noise will be always there. For human brain (e.g. for audio signals) – anything beyond 16 bits for each sample, the associated quantization noise becomes imperceptible.

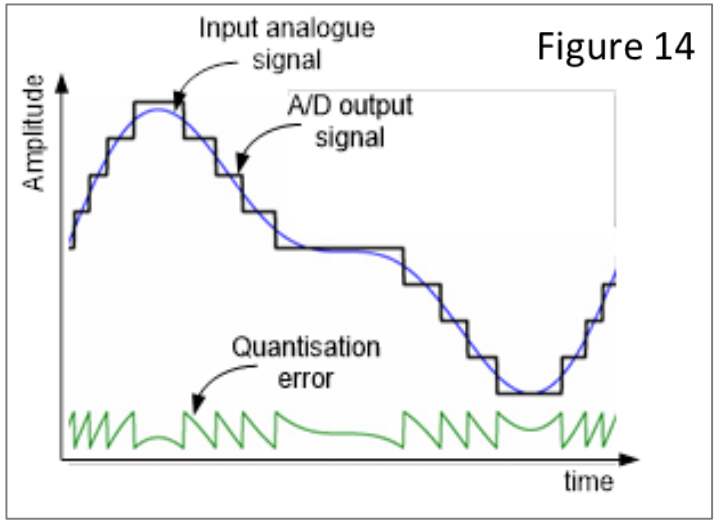


Figure 14

Quantisation (error) noise is random and uniformly distributed with zero mean.

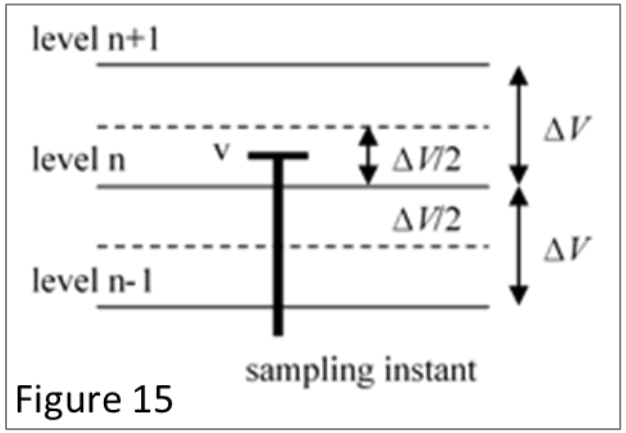


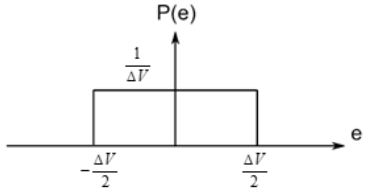
Figure 15

Maximum Quantisation error (e[n]) = $\frac{\Delta V}{2}$

Quantisation errors

Quantisation error (e)

- The probability density function of the error $P(e)$ has the form as shown below:



- The quantisation noise power or variance $\sigma_e{}^2$ is hence given by:

$$\sigma_e^2 = \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 P(e)de$$

$$= \frac{1}{\Delta V}\int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 de = \frac{1}{\Delta V}\left[\frac{e^3}{3}\right]_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} = \frac{\Delta V^2}{12}$$

Note: Probability of quantisation error is constant:

$P(e)=1/\Delta V$

- For uniform quantisation [all steps (ΔV) are of equal size], the quantisation noise power:

$\sigma_e^2 = \frac{\Delta V^2}{12}$ {**rms** quantisation error $\boldsymbol{\sigma_e}= \Delta/\sqrt{12}$ }

Signal-to-Quantisation ratio (SQNR)

- Signal-to-quantisation noise power ratio (SQNR) is defined as:

$$SQNR = \frac{P_s\ (signal\ power)}{P_n\ (noise\ power)}$$

$$P_s = \frac{1}{N}\sum_{n=1}^{N}\{x[n]\}^2; \quad P_n = \frac{1}{N}\sum_{n=1}^{N}\{e[n]\}^2; \quad P_n = \sigma_e^2 = \frac{\Delta V^2}{12}$$

$$\therefore SQNR(dB) = 10\,log\frac{P_s}{P_n} = 10\,log\frac{\sum_{n=1}^{N} x^2[n]}{\sum_{n=1}^{N} e^2[n]}$$

- The dynamic range, R, of the signal is defined as

$$R = \{x[n]\}_{\max} - \{x[n]\}_{\min}$$

- The quantisation step size of resolution is defined as $\Delta V = \frac{R}{L}$ (L- number of levels in the quantiser).

$$\therefore SQNR(dB) = 10\,log\frac{P_s}{\frac{(R/L)^2}{12}} = 10\,log\frac{12P_sL^2}{R^2}$$

$$SQNR(dB) = 10\,log\,P_s + 20\,log\,L + 10\,log\,12 - 20\,\log R$$

Quantisation errors

SQNR

- For the sine wave input, the average signal power is $A^2/2$, i.e. $(A/\sqrt{2})^2$ rms value. The signal-to-quantisation noise power ratio (SQNR) in decibels is:

$$SQNR = 10\,log\left(\frac{\frac{A^2}{2}}{\frac{\Delta V^2}{12}}\right) = 10\,log\left(\frac{\frac{A^2}{2}}{\frac{(2A/2^B)^2}{12}}\right) = 10\,log\left(\frac{3\times 2^{2B}}{2}\right)$$

$$\boldsymbol{SQNR = 6.02B + 1.76 \quad dB}$$

- The SQNR increases with the number of bits, *B*. In many DSP applications, an A/D converter resolution between 12 and 16 bits is adequate.
- It can be seen from the table below that the signal-to-quantisation noise ratio increases approximately 6dB for each bit.

Number of bits	Number of Levels	SQNR
3	8	19.7 dB
4	16	25.3 dB
5	32	31.6 dB
6	64	37.7 dB
7	128	43.8 dB

Example 18:

- A sampled signal that varies between -2V and 2V is quantised using B bits. What value of B will ensure that the quantisation noise power is less than 25×10^{-6}?

$$\text{Resolution } (\Delta V) = \frac{R \text{ (Dynamic range)}}{L \text{ (No of levels in the quantiser)}}$$

$\sigma_e^2 = \frac{\Delta V^2}{12} = 25 \times 10^{-6}$; $\therefore\ \Delta V^2 = 12 \times 25 \times 10^{-6}$

R = 2 – (-2) =4; $\therefore \Delta V^2 = \left(\frac{R}{L}\right)^2 = \frac{16}{L^2}$

$\frac{16}{L^2} = 12 \times 25 \times 10^{-6}$; L = 230.94

$L = 2^B - 1$; $\therefore 2^B = 231.94$

B = 7.86; Number of bits = 8;

Digital to Analogue Conversion (D/A) – Signal recovery

D/A converter

- The D/A conversion process is employed to convert the digital signal into analogue form, after it has been digitally processed.
- The reason for this conversion may be, for example, to generate an audio signal to drive a loudspeaker, or to sound an alarm. The D/A process is shown in Figure 16.
- A register is used to buffer the D/A's input to ensure that its output remains the same until the D/A is fed with the next digital input.
- The inputs to the D/A are a series of impulses, while the output of the DAC has a staircase shape as each impulse is held for a duration of *T* sec.
- The D/A shown in Figure 16 is referred to as a zero-order hold (i.e. holding each sample value for one sample interval)
- By comparing its output and its input y[n], it is evident that for each digital code fed into the D/A, its output is held for a duration of *T*. The result is the characteristic staircase shape at the D/A output.
- The D/A output approximates the analogue signal by a series of rectangular pulses, whose heights are equal to the corresponding value of the signal pulse.

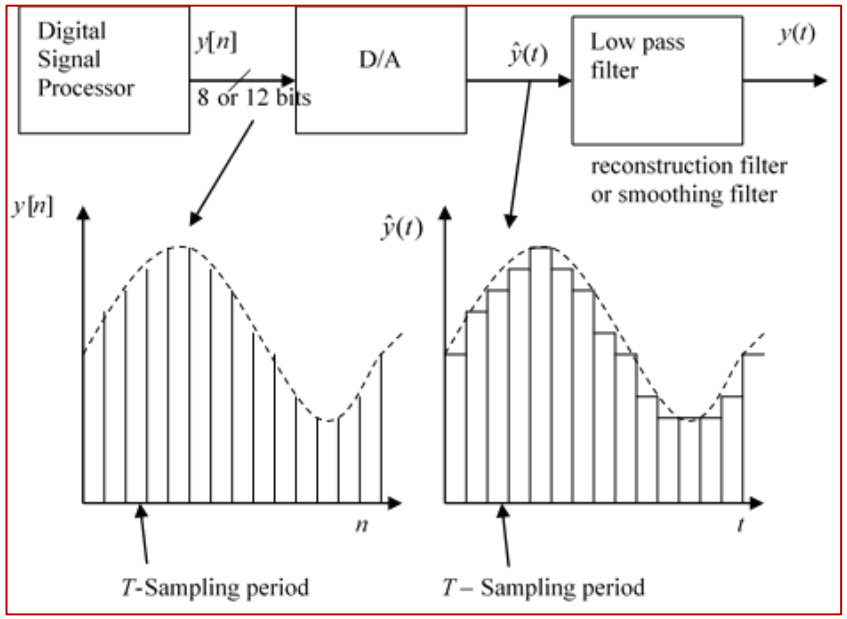


Figure 16: Conversion process from digital signals to analogue signals.

Reconstruction filter

The output of the D/A converter contains unwanted high frequency at multiples of the sampling frequency as well as the desired frequency components. The role of the output filter is to smooth out the steps in the D/A output thereby removing the unwanted high frequency components. In general, the requirements of the smoothing filter are similar to those of the anti-aliasing filter.

ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Discrete-time signals & difference equations (pp. 3-11)
- Time and frequency domain analysis (pp. 12-15)
- Sampling theorem & aliasing (pp. 16-23)
- A/D conversion & quantisation errors (pp. 24-31)
- Tutorial questions (pp. 32-36)

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2021

Topic 1: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

1. An analogue signal contains frequencies up to 10 kHz. Suppose that we sample this signal at a sampling frequency f_s = 8kHz. Examine what will happen to the frequency f_1 = 5 kHz? (Tick the correct answer)
   a) 5kHz frequency will remain unchanged
   b) 5kHz will alias to 3kHz
   c) 5kHz will alias to 0Hz
   d) None of the above

2. Which of the following is a not periodic signal ? (Tick the correct answer)
   a) x[n] = sin[πn/9]
   b) x[n] = sin[nπ]
   c) x[n] = cos[πn^2/15]
   d) x[n] = sin[πn/5 + π] + cos[πn/10 – π]

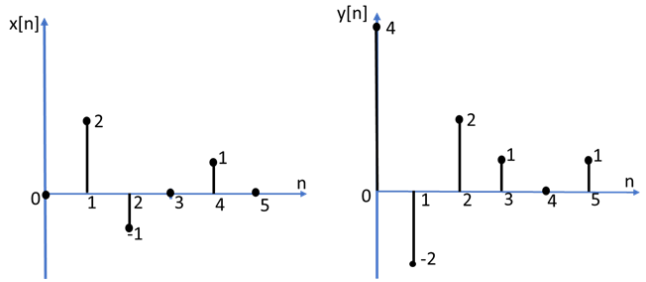
3. Given a real-valued bandpass signal x_a(t) with a bandpass spectrum of X_a(f) nonzero only for $9\ kHz\ < |f| < 12kHz$, determine the minimum sampling frequency (f_s) for which no aliasing occurs? (Tick the correct answer)
   a) f_S = 18 kHz
   b) f_S = 6 kHz
   c) f_S = 24 kHz
   d) None of the above

4. A discrete-time signal x(n) = 6.35 cos(nπ/10) is quantised with a resolution Δv=0.2 $volts$. How many bits are required in the Analogue to Digital Converter? (Tick the correct answer)
   a) 6 bits
   b) 8 bits
   c) 12 bits

SOLO – Level 1

5. Consider the amplitude modulated signal $x(t) = 5\,cos(3\pi t)\,cos(\pi t)$. Determine its spectral frequency components [you may assume that the frequencies are in kHz) (Tick the correct answer)
   a) $\pm$1.5kHz and $\pm$0.5kHz
   b) $\pm$3kHz and $\pm$1kHz
   c) $\pm$2kHz
   d) $\pm$2kHz and $\pm$1kHz

6. Given the discrete-time sequences x[n] and y[n] shown, which difference formula best describes their relationship ? (Tick the correct answer)



   a) y[n] = x[n-1] + 2x[n+1]
   b) y[n] = 2x[n] – x[n-1]
   c) y[n] = 2x[n-1] + x[n-2]
   d) y[n] = 2x[n-2] – x[n]

7. Determine the period of the discrete time signal $x[n] = \sin(0.2n + \pi)$. (Tick the correct answer)

   (a) 1 sample (b) 5 samples
   (c) π samples (d) Non of the above

Topic 1: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

1. A sinusoidal signal with peak-to-peak amplitude of 5V is sampled at 50kHz with uniform quantisation. Find the minimum number of bits for the analogue to digital converter to achieve a SQNR of at least 92dB. State any assumptions made.
2. Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by

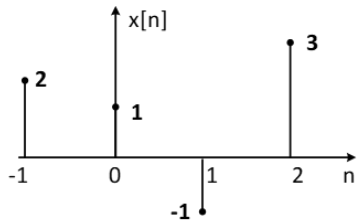
$$SQNR = 6.02B + 4.77 - 20\log\left({}^{A}/_{\sigma_{sig}}\right)$$

where the input range of the A/D converter is $\pm$A and the rms value of the input signal is σ_{sig}. Determine the SQNR if *B* is 16 bits and the input is a signal with a rms value of A/5.

3. An analogue signal $x_a(t) = \cos(2\pi 500t)$ is sampled at a rate of f_s=4kHz. Determine the resulting analogue and digital magnitude spectra.
4. An analogue signal $x(t) = \sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per sec:
   (a) Determine the Nyquist sampling rate for x(t).
   (b) Determine the folding frequency (or half the sampling frequency).
   (c) What are the frequencies, in radians, in the resulting discrete time signal x[n]?
   (d) If x[n] is passed through an ideal D/A converter what is the reconstructed signal y(t)?
5. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of $\pm$5 Volts is used to convert sampled voltages to integer codes.
   (a) Calculate the quantization interval of the A/D converter.
   (b) Determine the maximum quantization error.

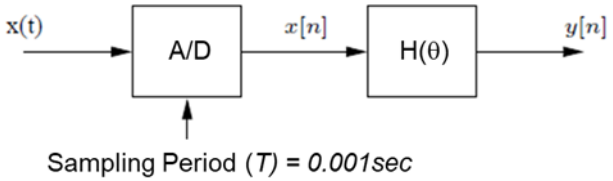
SOLO – Level 2

6. Consider the discrete-time sequence x[n] = cos[(nπ/8) + (π/4)] .
   (i) Determine the fundamental period of x[n].
   (ii) If the sampling frequency f_s = 10 kHz, write the equation of the analogue signal x(t).
7. Sketch the following:
   (c) x[n] = δ[n+2] + 2 δ[n-1] - 0.5 δ[n-3]
   (d) x[n] = u[-n-2]
   (e) x[n] = 0.5^n u[n-2]
8. Write the equation for x[n] (see diagram) as a sum of scaled and delayed unit impulses



9. For the system shown below , the input x(t) = cos(400πt) + cos(600πt). Find the output y[n] where

$$H(\theta) = \begin{cases} 1 & |\theta \le 0.5\pi| \\ 0 & otherwise \end{cases}$$



Topic 1: Tutorial questions (SOLO - Levels 3 & 4)

SOLO – Level 3

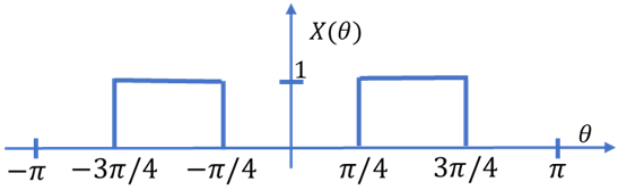
1. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ± 5 Volts is used to convert sampled voltages to integer codes. At what time (t>0) does the quantization error first reach its maximum value?
2. A system samples a signal x(t) = $\cos(\omega_o t)\cos(3\omega_o t)$ at f_s = 7 kHz and reconstructs the signal using an ideal LPF with a cutoff frequency of 3.5 kHz. The value of ω_o is $2\pi \times 10^3$ rad.
   (a) What is the highest frequency component present in the signal?
   Hint : $\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$
   (b) Sketch $|X(\omega)|$ and $|X(\theta)|$ – the magnitude spectrum of the signal before and after sampling for (a) above.
   (c) Can aliasing be prevented if the sampling frequency is increased to 8kHz? Explain.
3. If an analogue signal *x*(*t*) is sampled every *T* seconds, then at the output of the Analogue to Digital Converter, as shown in a diagram below. Using Inverse Fourier Transforms in Continuous time and discrete time domains, show that

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$



SOLO – Level 3

4. Find the Inverse DTFT of $X(\theta)$ for the figure shown below and show that $x[n] = \frac{2\sin(n\pi/4)\cos(n\pi/2)}{n\pi}$



$$X(\theta) = \sum_{n=-\infty}^{\infty} x(n)e^{-jn\theta}$$

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}d\theta$$

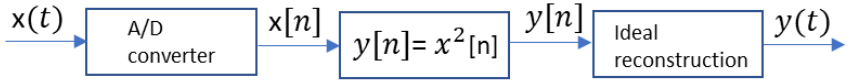
DTFT pair

θ = digital frequency; $-\pi \le \theta \le \pi$

5. Find the Fourier Transform of x(t) = δ(t-a). Hence, show that Hence, show that $\int_{-\infty}^{\infty} e^{j\omega t}dt = 2\pi\delta(\omega)$.

SOLO – Level 4

1. A speech signal is band limited to 4 kHz is ideally sampled at a sampling rate, f_s, and the sampled signal $x[n]$, is processed by squaring operations $y[n] = x^2$[n], whose output is ideally reconstructed to obtain y(t) as follows.



   What minimum sampling rate f_s, will ensure $y(t) = x^2(t)$?
2. The signal $x(t) = 2\cos(2000\pi t) - 4\sin(4000\pi t)$ is quantised by rounding, using a 12 bit quantiser. What is the rms quantisation error and the SQNR?

Topic 1: Tutorial questions (SOLO-Level 5)

SOLO – Level 5

1. Given a real-valued bandpass signal x_a(t) and $X_a(f) = 0$ for $|f| < f_1$ and $|f| > f_2$, the Nyquist sampling theorem says that the minimum sampling frequency is $f_s = 2f_2$. However , in some cases , the signal may be sampled at a lower rate.

   (a) Suppose that f_1 = 8 kHz and f_2 = 10 kHz. Draw a sketch of $X_a(\theta)$, if the sampling frequency f_s = 4 kHz.

   (b) If the bandwidth of the bandpass signal is defined as B = f_2 - f_1 and the centre frequency to be $f_c = \frac{(f_2+f_1)}{2}$, Show that if $f_c > {}^{B}/_{2}$ and f_2 is an integer multiple of the bandwidth B, no aliasing will occur if x_a(t) is sampled at a sampling frequency $f_s = 2B$.

   (c) Repeat part (b) for the case in which f_2 is not an integer multiple of the bandwidth B.

2. The D/A output approximates the analogue signal by a series of rectangular pulses, whose heights are equal to the corresponding value of the signal pulse as shown in Figure A.

   (a) Consider one pulse, $\hat{y}\ (t)$, with a height k and show that the corresponding frequency response $\hat{Y}\ (\omega)$ is a *sinc* function given by the following equation:

$$\hat{Y}(\omega) = k\,\frac{T}{2}\,e^{-\frac{j\omega T}{2}}\,\frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}}$$

   (b) Explore the effect of $\hat{Y}(\omega)$ on the output signal spectrum $Y(\omega)$ and illustrate your understanding of this using a spectral plot. If compensation is required for the $sinx/x$, roll-off inherent in the D/A process, what function would you use to compensate? Briefly explain why.

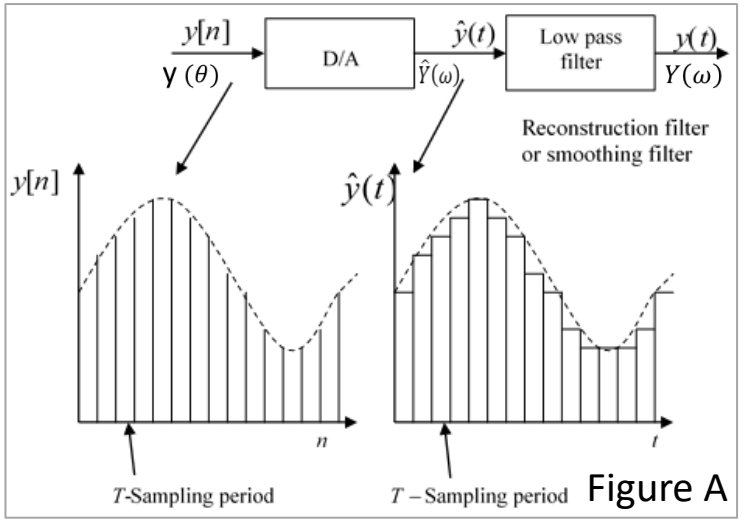


Figure A